

Identifying High-Value Locations and Best-Selling Products

An Analysis of Coffee Shop Sales Data

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```
library(tidyverse)
```

Warning: package 'ggplot2' was built under R version 4.5.2

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    4.0.2      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.1.0
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(ggplot2)
library(kableExtra)
```

Attaching package: 'kableExtra'

The following object is masked from 'package:dplyr':

group_rows

```
library(lubridate)
coffee_data <- read_csv("Data/coffee-shop-sales.csv")
```

Rows: 149116 Columns: 11

-- Column specification -----

Delimiter: ","

chr (5): transaction_date, store_location, product_category, product_type, ...

dbl (5): transaction_id, transaction_qty, store_id, product_id, unit_price

time (1): transaction_time

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

0.1 Executive Summary

0.2 Introduction

The central problem addressed in this report is how a new café can choose a location and initial product offering that maximises early sales and customer engagement. The report focuses on two key objectives: identifying the location with the strongest potential for expansion, and determining which product should be prioritised at launch. Together, these findings will support strategic decision-making for launching a new café with the highest likelihood of early success.

0.3 Dataset Introduction

To tackle this problem, the analysis uses the Coffee Shop Sales dataset from Maven Analytics, which provides detailed transaction-level information across multiple existing stores. The

dataset includes 149116 rows and 11 columns, representing a large volume of customer purchase activity. Key variables include numeric identifiers such as `transaction_id`, `store_id`, and `product_id`, which uniquely identify each transaction, store, and product. Temporal variables include `transaction_date` (stored as character) and `transaction_time` (stored as a time object), capturing when each purchase occurred. Measures of customer behaviour are represented through `transaction_qty`, a numeric variable indicating the number of items purchased per transaction. Categorical variables (stored as character) such as `store_location`, `product_category`, `product_type`, and `product_detail` describe where the transaction took place and the type of product purchased. Together, these variables provide a rich foundation for analysing sales patterns across locations and product categories.

0.4 Methodology

Before the data is used for analysis, a data cleaning process is performed. All variables are checked for missing values and no missing values are found across the dataset, thus no further action was taken. Next, the class of each variable is verified. The `transaction_date` variable, which contains date information, was stored as a character string and was therefore mutated to a date format to allow for further analysis.

Following the goal of determining the best location and best selling products, the relationship between product category, product details, sales quantity, unit price, and store location is explored.

0.4.1 Summary of data - Sales by location

Table 1: Total sales and quantity sold by store location

Store Location	Quantity Sold	Average Price	Total Sales
Hell's Kitchen	71737	3.30	236511.2
Astoria	70991	3.27	232243.9
Lower Manhattan	71742	3.21	230057.2

Table 1 presents the total quantity sold, average product price, and total sales of stores in each location. The store in Hell's Kitchen recorded the highest total sales amount among the three locations, despite having a slightly lower sales quantity compared to the store in Lower Manhattan. Nevertheless, the slightly higher average price contributed to the higher total sales figure.

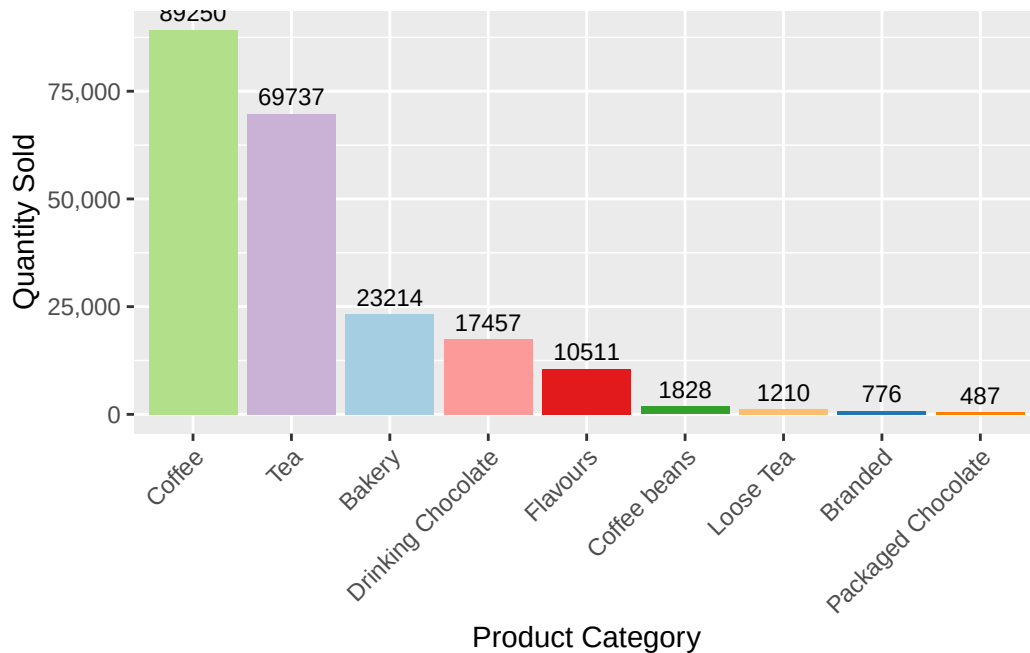


Figure 1: Sales by product category

0.4.2 Summary of data - Sales by product category

Figure 1 presents the quantity sold by product category. Coffee is the best selling product category, followed by tea. The third best selling category is bakery items; however, the difference between tea and bakery sales is significant, as bakery sold less than half of total tea sold, representing a large gap between the top two best selling categories and the rest.

1 Results

```
# clean and prep data
coffee_sales_clean <- coffee_data %>%
  mutate(sales = transaction_qty * unit_price)
head(coffee_sales_clean)
```

```
# A tibble: 6 x 12
  transaction_id transaction_date transaction_time transaction_qty store_id
      <dbl>      <date>           <time>           <dbl>      <dbl>
1             1 2023-01-01      07:06:11             2          5
```

Table 2: Summary statistics by store location in New York City.

```
# overall store performance
location_summary <- coffee_sales_clean %>%
  group_by(store_location) %>%
  summarise(total_sales = sum(sales),
            total_transactions = n(),
            total_quantity_sold = sum(transaction_qty),
            avg_transaction_value = mean(sales, na.rm = TRUE),
            avg_daily_sales = total_sales / n_distinct(transaction_date)) %>%
  arrange(desc(total_sales))
```

```
location_summary
```

```
# A tibble: 3 x 6
  store_location total_sales total_transactions total_quantity_sold
  <chr>          <dbl>          <int>          <dbl>
1 Hell's Kitchen 236511.          50735          71737
2 Astoria        232244.          50599          70991
3 Lower Manhattan 230057.          47782          71742
# i 2 more variables: avg_transaction_value <dbl>, avg_daily_sales <dbl>
```

```
2      2 2023-01-01      07:08:56      2      5
3      3 2023-01-01      07:14:04      2      5
4      4 2023-01-01      07:20:24      1      5
5      5 2023-01-01      07:22:41      2      5
6      6 2023-01-01      07:22:41      1      5
# i 7 more variables: store_location <chr>, product_id <dbl>, unit_price <dbl>,
#   product_category <chr>, product_type <chr>, product_detail <chr>,
#   sales <dbl>
```

1.1 Which Location would be Most Suitable based on Sales?

Table 2 shows the total sales, total transactions, total quantity sold, average transaction value, and average daily sales of the three respective locations in NYC.

```
ggplot(location_summary,
  aes(x = reorder(store_location, total_sales), y = total_sales)) +
  geom_col(fill = "brown") +
  coord_flip() +
```

```
labs(title = "Total Sales by Store Location in NYC", x = "Store Location", y = "Total Sales")  
theme_minimal()
```

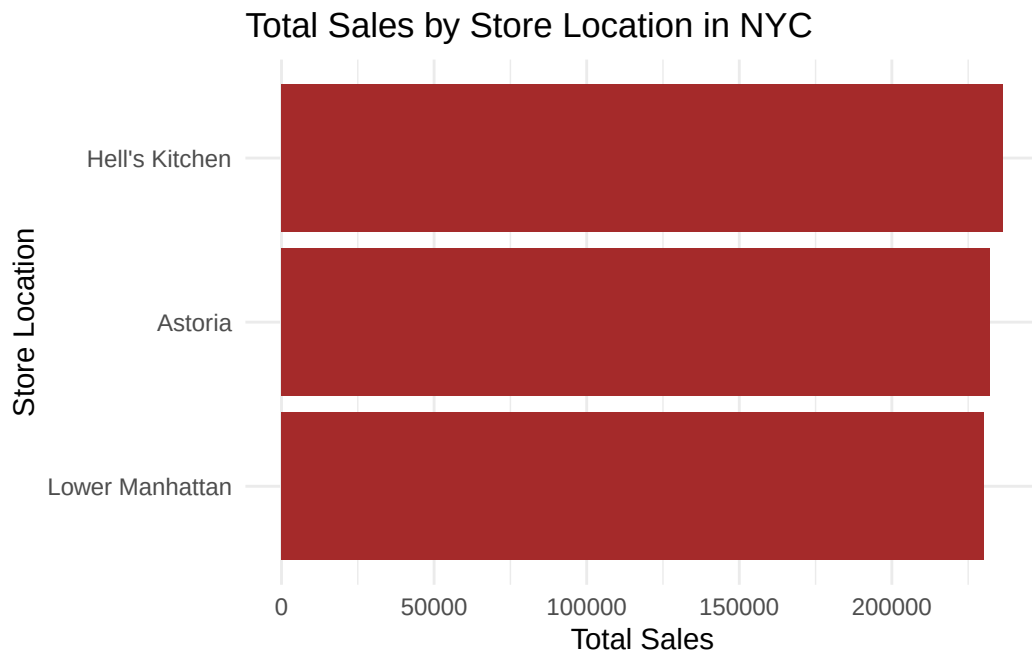


Figure 2: Total sales by store location in New York City.

As shown in Figure 2, **Hell's Kitchen** generated the highest total sales overall.

1.2 Optimal Menu Recommendations

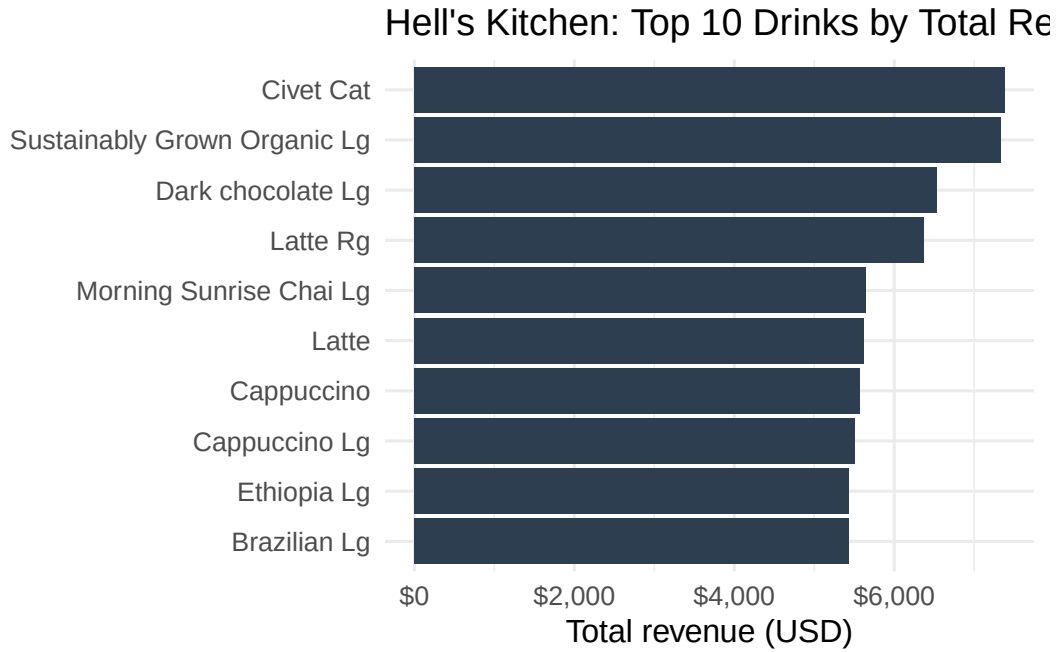


Figure 3: top 10 drinks by total revenue (hell's kitchen)

As shown in Figure 3, total revenue in Hell's Kitchen is concentrated among a small set of drinks, with a clear drop-off after the top-ranked items. The highest-earning products include premium coffees and larger-size beverages, suggesting that both higher unit prices and steady demand contribute to revenue. This figure highlights the key drivers of takings and provides an evidence-based starting point for deciding which drinks should be prioritised for menu placement and promotion.

Table 3: top 10 drinks by total revenue (hell's kitchen)

drink	total_quantity	total_revenue
Civet Cat	164	7380.00
Sustainably Grown Organic Lg	1543	7329.25
Dark chocolate Lg	1452	6534.00
Latte Rg	1498	6366.50
Morning Sunrise Chai Lg	1413	5652.00
Latte	1500	5625.00
Cappuccino	1487	5576.25
Cappuccino Lg	1297	5512.25

Table 3: top 10 drinks by total revenue (hell’s kitchen)

drink	total_quantity	total_revenue
Brazilian Lg	1552	5432.00
Ethiopia Lg	1552	5432.00

Table 4: top 10 drinks by total quantity sold (hell’s kitchen)

drink	total_quantity	total_revenue
Ouro Brasileiro shot	1854	5056.20
Serenity Green Tea Rg	1601	4002.50
Morning Sunrise Chai Rg	1589	3972.50
Brazilian Lg	1552	5432.00
Ethiopia Lg	1552	5432.00
Our Old Time Diner Blend Sm	1550	3100.00
Sustainably Grown Organic Lg	1543	7329.25
English Breakfast Lg	1529	4587.00
Earl Grey Rg	1522	3805.00
Brazilian Sm	1520	3344.00

Tables Table 3 and Table 4 together identify the best menu candidates by combining profitability and demand. Items near the top of both tables are strong “core menu” choices because they sell frequently and generate high total revenue, making them reliable for day-to-day operations. Drinks that rank highly in Table 3 but not Table 4 likely achieve revenue through a higher price point; these are good premium offerings to keep for margin and upselling, but they may require targeted promotion and careful stock planning. Conversely, drinks that are very popular in Table 4 but lower in Table 3 are dependable volume sellers and can be used as anchors for bundles or add-ons. Overall, a balanced menu should prioritise overlapping winners, then supplement with a small set of premium specials.